

PIONEER JUNIOR COLLEGE, SINGAPORE

JC1 PRELIMINARY EXAMINATIONS
HIGHER 1

CANDIDATE
NAME

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INDEX
NUMBER

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CHEMISTRY

8873/02

Paper 2 Structured

10 September 2018

2 hours

Candidates answer on the Question Paper.
Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** the questions.

Section B

Answer only **one** question.

You are advised to show all working in calculations.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

FOR EXAMINER'S USE					
Paper 1		Paper 2 Section A		Paper 2 Section B	
Total	/ 30	4	/ 10	Total	/ 20
		5	/ 6		
Paper 2 Section A		6	/ 7	Penalty	s.f. / units
1	/ 4	7	/ 5	TOTAL	/ 110
2	/ 9	8	/ 8	GRADE	
3	/ 11	TOTAL	/ 60		

Answer **all** the questions in this section, in the spaces provided.

- 1 Fig. 1.1 shows a sketch of the logarithm of the first fourteen ionisation energies, $\log(\text{IE})$, of element **A** against the number of electrons removed.

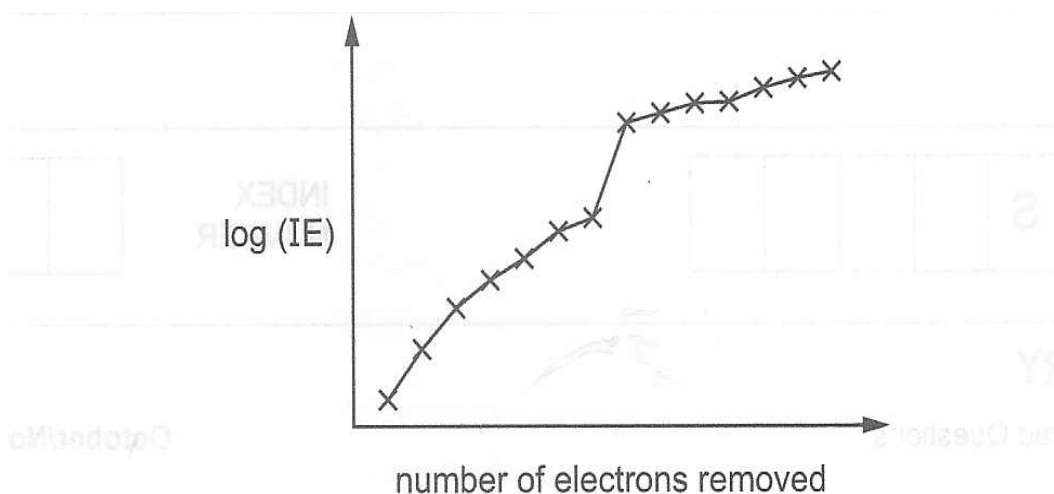


Fig. 1.1

- (a) Write an equation for the second ionisation energy of element **A**. [1]



- (b) Explain the following features of Fig. 1.1.

- (i) The increase in ionisation energies as the electrons are removed. [1]

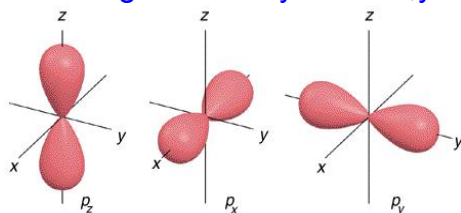
Ionisation energies increases generally because electrons are removed from an increasingly positive charged gaseous ion.

- (ii) The significant jump in values from the 7th to the 8th ionisation energy. [2]

The eighth electron to be removed comes from the inner quantum shell, which is nearer to and less shielded from the nucleus, hence stronger attractive forces by the nucleus

- (c) Sketch the shape of the orbital of element A that is not fully filled. [1]

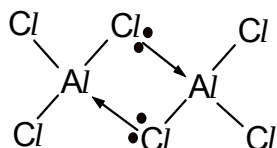
Students to sketch the p orbital aligned on any of the x,y or z axis.



[Total: 4]

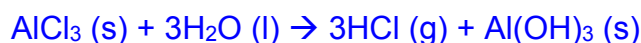
- 2 When dry chlorine is passed over heated aluminium foil in a hard glass tube, a vapour is produced which condenses to a yellow-white solid in the cooler parts of the tube. At low temperatures the vapour has the empirical formula AlCl_3 and an M_r of 267.

- (a) Suggest the molecular formula of the vapour, and draw a structure to describe its bonding. [2]

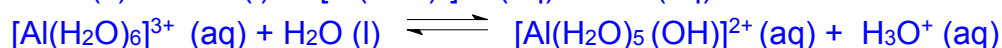


The yellow-white solid reacts with water in two separate ways.

- When a few drops of water are added to the solid, steamy white fumes are evolved and a white solid remains, which is insoluble in water.
 - When a large amount of water is added to the solid, a clear, weakly acidic solution results
- (b) Write equations, including state symbols, for these two reactions and explain the observations. [4]



AlCl_3 reacts with small amount of water to give white steamy fumes of HCl and white solid of $\text{Al}(\text{OH})_3$.



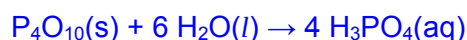
Hydrolysis occurs and H_3O^+ is produced which makes the solution acidic.

- (c) Describe and explain what you would see when a sample of $\text{Na}_2\text{O}(\text{s})$ and P_4O_{10} is added to a solution of litmus in water separately. Write equations for the reaction that occur. [3]

Note: The litmus solution is a purple solution when it is added to a neutral solution.



The litmus solution changed from purple to blue.



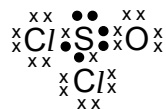
The litmus solution changed from purple to red.

[Total: 9]

- 3 (a) In a molecule of SOCl_2 , the sulfur atom has four bonds.

Draw a 'dot and cross' diagram of SOCl_2 .

[1]



- (b) When SOCl_2 is reacted with a carboxylic acid to produce an acyl chloride, two acidic gases are formed.



A 1.00g sample of a carboxylic acid RCOOH was treated in this way, and the gases were absorbed in 60.0cm^3 of 0.500 mol dm^{-3} $\text{NaOH}(\text{aq})$, in excess.

- (i) Write equations for the reactions between NaOH and SO_2 . [1]



The excess NaOH was titrated with 0.500 mol dm^{-3} $\text{H}^+(\text{aq})$. It required 10.8 cm^3 of the $\text{H}^+(\text{aq})$ solution to reach the end-point.

- (ii) Calculate the total amount, in moles of NaOH that reacted with the SO_2 and HCl . [2]

$$\text{Initial amt of NaOH} = 0.5 \times 60 / 1000 = 3.00 \times 10^{-2} \text{ mol}$$

$$\text{Amt of NaOH unreacted} = 0.5 \times 10.8 / 1000 = 5.40 \times 10^{-3} \text{ mol}$$

$$\text{Amt of NaOH reacted} = (3.00 - 0.54) \times 10^{-2} = 2.46 \times 10^{-2} \text{ mol}$$

- (iii) Calculate the amount, in moles of RCOOH that produced the SO_2 and HCl . [1]

$$\text{Amt of RCOOH} = (2.46 \times 10^{-2}) / 3 = 8.20 \times 10^{-3} \text{ mol}$$

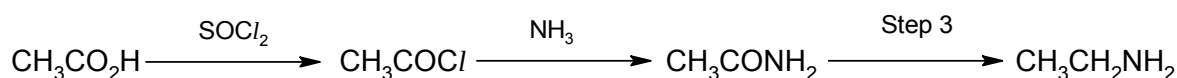
- (iv) Hence calculate the M_r of the carboxylic acid, RCOOH . [1]

$$M_r = 1.00 / (8.20 \times 10^{-3}) = 121.95 (=122)$$

- (v) The R group contains carbon and hydrogen only. Suggest the molecular formula of RCOOH . [1]



- (c) The following synthetic route shows how a carboxylic acid can be converted into an amine.



(i) Suggest a reagent for step 3.

[1]

LiAlH_4 or $\text{H}_2(\text{g})$

Angelic acid, $\text{C}_5\text{H}_8\text{O}_2$, is a natural product isolated from the roots of the angelica plant.

- Angelic acid reacts with $\text{H}_2 + \text{Ni}$ to form **T**, $\text{C}_5\text{H}_{10}\text{O}_2$.
- **T** undergoes the above synthetic route to form the amine **U**, $\text{C}_5\text{H}_{13}\text{N}$.
- **U** can also be made by reacting 1-bromo-2-methylbutane with ammonia.

Angelic acid displays *cis-trans* isomerism.

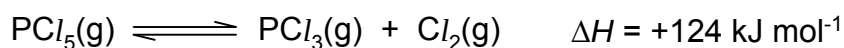
(ii) Suggest the structures for angelic acid, **T** and **U**.

[3]

$\text{CH}_3\text{CH}=\text{C}(\text{CH}_3)\text{COOH}$	$\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{COOH}$	$\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{NH}_2$
Angelic acid	T	U

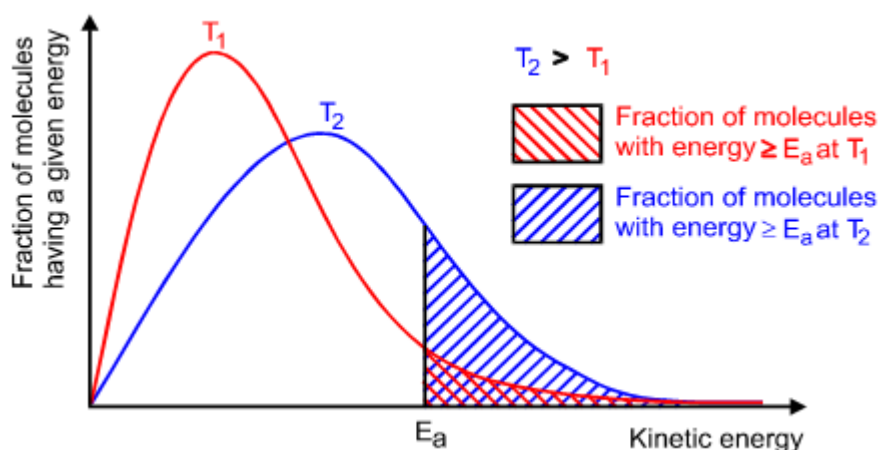
[Total: 11]

- 4 At 450K phosphorus(V) chloride, $\text{PCl}_5(\text{g})$ decomposes to form phosphorus(III) chloride, $\text{PCl}_3(\text{g})$, and chlorine, $\text{Cl}_2(\text{g})$. A dynamic equilibrium is established as shown.



- (a) (i) With the use of a Boltzmann distribution curve, explain why an increase in temperature will increase the rate of decomposition of $\text{PCl}_5(\text{g})$.

[4]



When temperature is increased, there is:

- an increase in the average kinetic energy of the particles, resulting in,

- an increase in frequency of collision
- an increase in the number of reacting particles with energy $\geq E_a$
- an increase in the frequency of effective collisions between reacting particles

Boltzmann distribution curve

- correct axis,
- legend
- the graph drawn correctly
(assymetrical shape, T_2 peak shifted right and downwards),
- indicating the E_a as cut off for area under curve.

- (ii) State and explain the effect of increasing temperature on the percentage of $PCl_5(g)$ that decomposes. [2]

Increase in temperature will always favour endothermic reaction. Equilibrium shifts to the right / forward endothermic reaction is favoured to absorb some of the added heat. The percentage decomposition of $PCl_5(g)$ will increase.

- (b) Explain the meaning of the term dynamic equilibrium and the conditions necessary for it to become established. [2]

Dynamic equilibrium: When the rates of forward and reverse / backward reactions are equal

Conditions: closed / sealed system/container

- (c) When 2.00 mol of $PCl_5(g)$ are decomposed at 450K and 1.00×10^5 Pa in a 7.5 dm^3 container, the resulting equilibrium mixture contains 0.800 mol of $Cl_2(g)$.

Calculate the value of K_c and state its units. [2]

	$PCl_5(g)$	$\rightleftharpoons$	$PCl_3(g)$	+	$Cl_2(g)$
Initial amount / mol	2.00		0		0
Change in amount / mol	-0.800		+0.800		+0.800
Equilibrium amount / mol	1.20		0.800		0.800
Equilibrium conc / mol dm^{-3}	$\frac{1.2}{7.5} = 0.160$		$\frac{0.8}{7.5} = 0.1067$		$\frac{0.8}{7.5} = 0.1067$

$$K_c = \frac{(0.1067)^2}{(0.160)} = 0.0711 \text{ mol dm}^{-3}$$

[Total: 10]

- 5 (a) Magnesium(I) chloride, $MgCl$, is an unstable compound and readily decomposes as shown.



Use the following data to calculate the enthalpy change of this reaction.

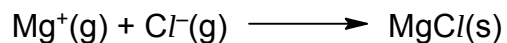
[1]

$$\Delta H_f \text{MgCl(s)} = -106 \text{ kJ mol}^{-1};$$

$$\Delta H_f \text{MgCl}_2\text{(s)} = -642 \text{ kJ mol}^{-1}$$

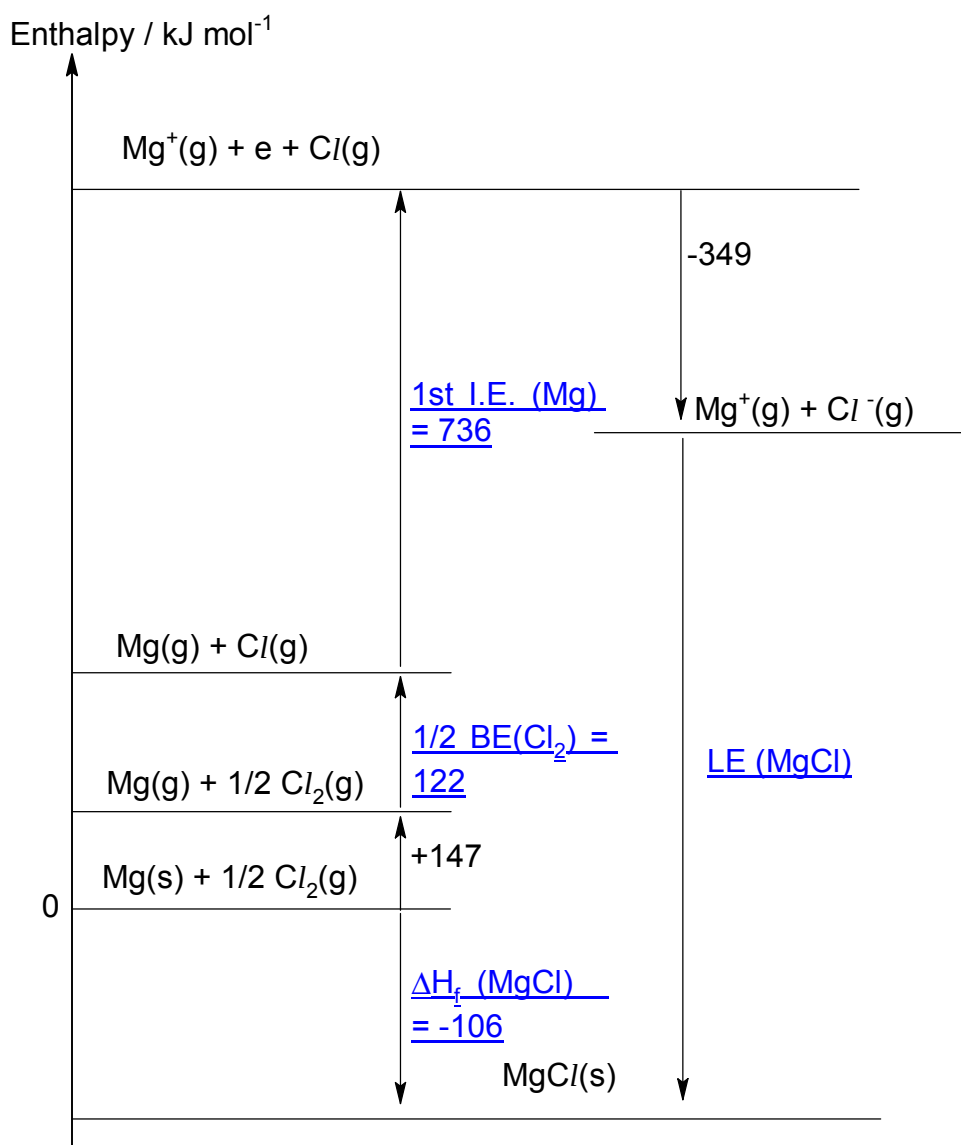
$$\text{enthalpy change} = (-642) - (2 \times -106) = -430 \text{ kJ mol}^{-1}$$

- (b) (i) The equation for which ΔH is the lattice energy for MgCl is shown.



Label the relevant enthalpy changes in the energy cycle below.

[2]



- (ii) Use the energy cycle above and relevant data from the *Data Booklet* to calculate a value for the lattice energy of MgCl . [1]

By Hess' Law,

$$-106 = 147 + 122 + 736 + (-349) + \text{LE}(\text{MgCl})$$

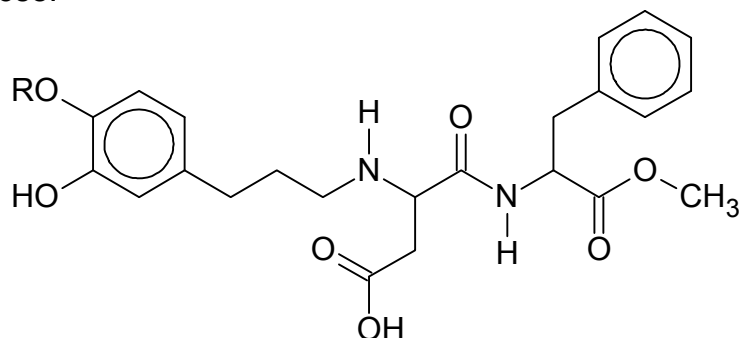
$$\text{LE}(\text{MgCl}) = -762 \text{ kJ mol}^{-1}$$

- (iii) Suggest how the lattice energy of MgCl_2 will compare to that of MgCl . Explain your answers. [2]

The lattice energy of MgCl_2 is more exothermic / more negative than that of MgCl . This is due to higher charge and smaller radius of Mg^{2+} cation compared to that of Mg^+ .

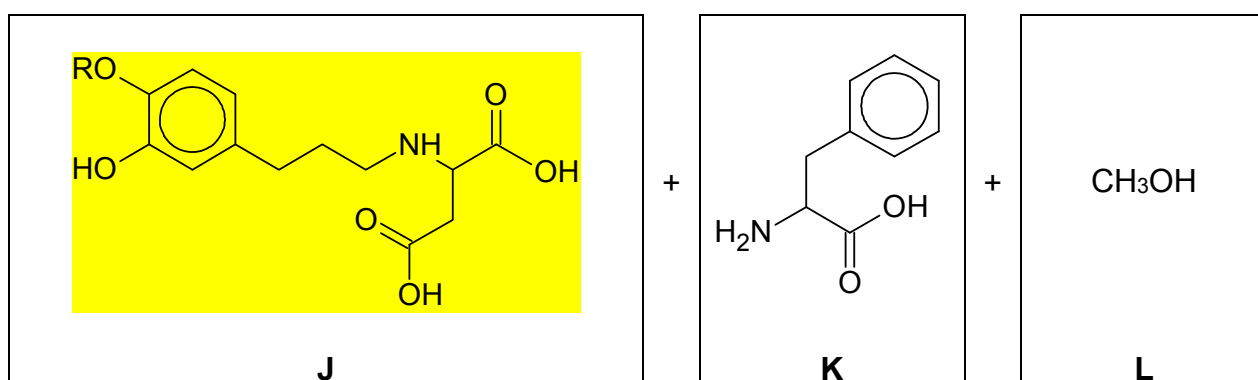
[Total: 6]

- 6 The compound *Advantame* is a sweetener that tastes approximately 25000 times sweeter than sucrose.



Advantame

- (a) The decomposition of *Advantame* produces three molecules, **J**, **K** and **L**. The RO-group in *Advantame* is unreactive.



- (i) Name the type of reaction occurring. [1]

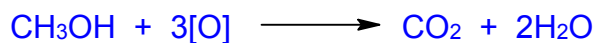
Hydrolysis

- (ii) Draw the structure of **J** in the box above. [1]

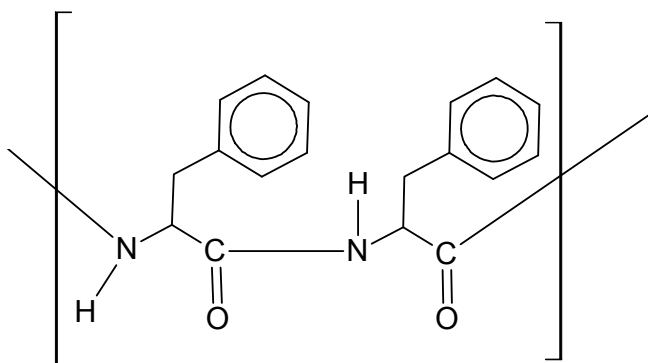
- (b) (i) State what you would observe when acidified potassium manganate(VII) is added to a solution of **L**. [2]

Purple potassium manganate(VII) decolourised. Carbon dioxide gas evolved which gives a white ppt with limewater.

- (ii) Write balanced equation for the reaction taking place in (b)(i). [1]

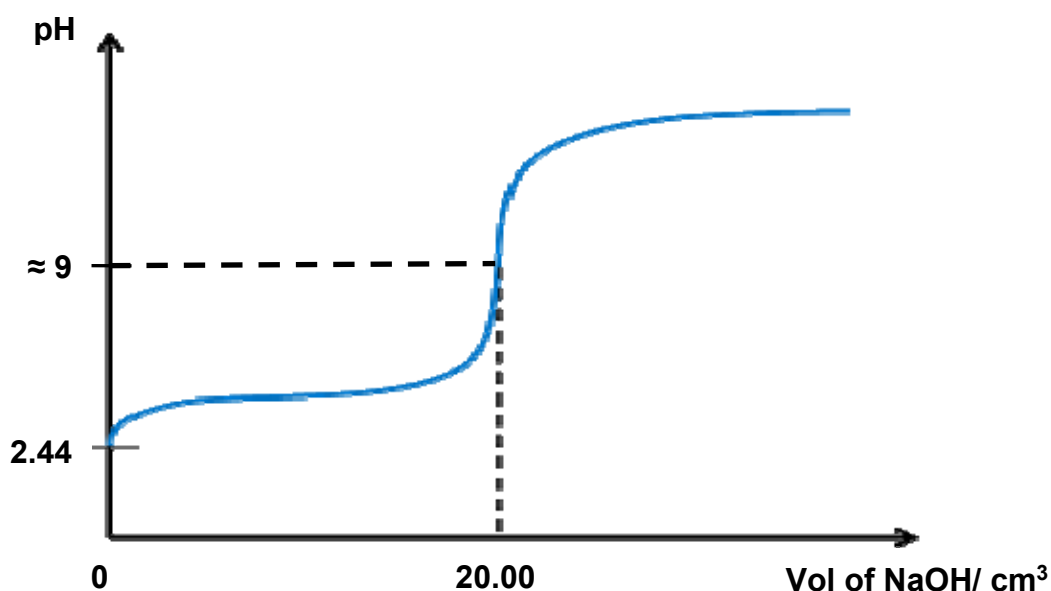


- (c) **K** can be polymerised. Draw the structure of the polymer showing two repeat units. The linkage between the monomer units should be fully displayed. [2]

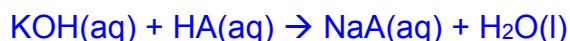


[Total: 7]

- 7 A 25.0 cm³ of 0.72 mol dm⁻³ of CH₃COOH in a conical flask was titrated against aqueous sodium hydroxide and the pH of the solution in the conical flask was plotted against volume of NaOH(aq).



- (a) (i) Using the graph, determine the concentration of aqueous sodium hydroxide in the burette. [1]



$$n(\text{KOH}) = n(\text{HA}) = \frac{25.0}{1000} \times 0.72$$

$$[\text{KOH}] = \frac{\frac{25.0}{1000} \times 0.72}{\frac{20.00}{1000}} = 0.90 \text{ mol dm}^{-3}$$

- (ii) Using the data in the graph, show that ethanoic acid is a weak acid. [1]

$$[\text{H}^+]_{\text{eqm}} = 10^{-2.44} = 3.63 \times 10^{-3} \text{ mol dm}^{-3} < [\text{CH}_3\text{COOH}]_{\text{initial}} = 0.72 \text{ mol dm}^{-3}$$

hence it does not dissociate completely

- (b) Circle the region on the graph on this question paper where a buffer solution is produced from the titration. [1]

Circle the region before equivalence point and after 0 cm³ of NaOH(aq) added.

- (c) Phenolphthalein indicator is used in this titration. Using the information in the table provided below, give two reasons why phenolphthalein is a good indicator. [2]

Indicator	pH range	Colour change	
		Colour in acid < pH 8.2	Colour in alkali > 10.0
Phenolphthalein	8.2 – 10.0	Colourless	Red

The equivalence point of pH ≈ 9 coincides within the phenolphthalein working pH range from pH = 8.2 – 10.0

or

The pH working range from pH 8.2-10.0 coincides within the rapid change of pH of the titration curve

And

There is a distinct colour change of the indicator from colourless to pink at the end point.

[Total: 5]

- 8 Many elements have *allotropes*. Carbon has a number of allotropes including diamond, graphite and graphene. Each of these allotropes has different properties. This means the allotropes can be used for different purposes.

Diamond is hard and does not conduct electricity.

Graphite is soft and has high electrical conductivity.

Graphene is a nanomaterial and has high electrical conductivity. A large layer of graphene would be very strong.

- (a) (i) Describe what is meant by allotropes. [1]

Different structures containing the same type of atoms.

- (ii) Describe the structure of graphene. Explain how this structure gives graphene the properties of high strength and high electrical conductivity.

You may find it useful to include a labelled diagram in your answer. [3]

A single layer of graphite (monoatomic layer of carbon), with sp^2 hybridised carbon atoms arranged in a hexagonal lattice and unhybridised 2p orbitals above and below the plane of the sheet.

Each carbon atom uses all three sp^2 orbitals to overlap head-on with the sp^2 orbitals of three adjacent carbon atoms, forming three carbon-carbon bonds. Because of the network of strong covalent bonds between the sp^2 hybridised carbon atoms, graphene has a very high tensile strength.

The unhybridised 2p orbitals of the carbon atoms overlap sideways with one another to form a delocalised pi electron cloud above and below the plane of carbon atoms.

The remaining electron in the unhybridised 2p orbitals per atom is delocalised over the whole graphene layer and responsible for the excellent electrical conductivity.

- (b) Research in nanotechnology is growing and many new nanomaterials are being produced with ever more uses. Uses of nanoparticles and nanomaterials is relatively new and there are concerns about their direct impact on human health and environment.

For example, nanoparticles are used to deliver drugs within cells.

- (i) State the difference between a nanoparticle and a nanomaterial in terms of size. [1]

Nanoparticles ALL dimensions 1–100nm/on the nanoscale
AND nanomaterial (at least) one dimension 1–100nm/on the nanoscale

- (ii) Suggest what properties of nanoparticles enable them to deliver drugs within cells. Explain your answer. [2]

Nanoparticles are smaller than (animal) cells or they can pass through the cell membrane or pass into/between cells
larger amount of Drugs can be EASILY bound to/enclosed by the nanoparticle due to large surface to volume ratio of the nanoparticle.

- (iii) Give one example of an impact of nanoparticles on the environment. [1]

Any valid answer

Use of photocatalytic copper tungsten oxide nanoparticles to break down oil into biodegradable compounds.

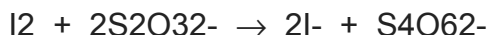
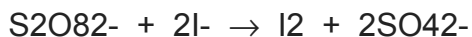
Use of gold nanoparticles embedded in a porous manganese oxide as a room temperature catalyst to breakdown volatile organic pollutants in air

[Total: 8]

Section B

Answer **one** question from this section, in the spaces provided.

- 1 If potassium iodide and ammonium peroxodisulfate are mixed in the presence of starch and sodium thiosulfate, the iodine liberated reacts with the thiosulfate ions until they are all used up.



When no thiosulfate ions are left, any iodine formed reacts with starch and the dark blue colour of the starch-iodine complex appears. The time taken for the dark blue colour to appear with a constant quantity of sodium thiosulfate present depends on the rate of formation of iodine

Table of results

Experiment	Volume of I^- / cm^3	Volume of $\text{S}_2\text{O}_8^{2-}$ / cm^3	Volume of deionised water / cm^3	Volume of starch / cm^3	t / s
1	10.00	20.00	0.0	1.0	23
2	10.00	10.00	10.0	1.0	46
3	10.00	10.00	10.0	1.0	40

- (a) (i) Using the results in your table, deduce the order of the reaction with respect to the peroxodisulfate ions. Explain clearly how you arrived at your answer. [2]

Comparing expt 1 and 2, keeping $[\text{I}^-]$ constant

Halve $[\text{S}_2\text{O}_8^{2-}]$, time is doubled, hence rate is halved, hence first order wrt $\text{S}_2\text{O}_8^{2-}$

Or substitution method

- (ii) Instead of washing the beaker as required before performing experiment 3, a student simply just poured away the reaction mixture. There were some left over reaction mixture in the beaker when he performed experiment 3. State and explain why the time is shorter than in experiment 2, assuming all other conditions are kept constant. [1]

The excess I_2 left in the flask will react with the $\text{S}_2\text{O}_3^{2-}$ in the new experiment. As there is less $\text{S}_2\text{O}_3^{2-}$ for the new experiment, less I_2 is reacted.

the time taken to produce lesser I_2 will be shorter than 46 seconds and hence observe the blue colour faster.

- (iii) Explain why a fixed amount of sodium thiosulfate is required. [1]

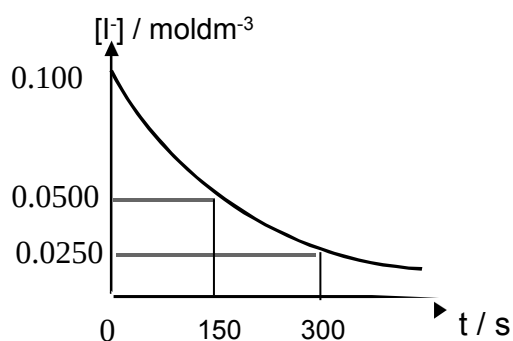
To ensure an equal amount of iodine is reacted before the solution turns dark blue in each experiment.

- (b) It was reported that the order of reaction with respect to iodide ions is one.

In order to substantiate this information, the chemistry teacher gives out these instructions to his class.

- 1) Prepare a solution mixture containing $0.100 \text{ mol dm}^{-3}$ potassium iodide and 1.00 mol dm^{-3} of ammonium peroxydisulfate.
- 2) Record the concentration of the iodide ions against time from the start of the reaction.
- 3) Plot a graph of concentration of iodide ions against time

- (i) Sketch the graph of concentration of iodide ions against time, given that the half life is 150 seconds, showing the values and important features of your graph to show that it is first order reaction with respect to iodide. [2]



- (ii) A student mistakenly prepared a solution containing 1.00 mol dm^{-3} potassium iodide and $0.100 \text{ mol dm}^{-3}$ of ammonium peroxydisulfate instead.

He recorded the concentration of the peroxydisulfate ions against time from the start of the reaction and plot a graph of concentration of peroxydisulfate ions against time on a piece of graph paper.

On realising the mistake, the teacher instructed him to carry another experiment by preparing a solution containing 2.00 mol dm^{-3} potassium iodide and $0.100 \text{ mol dm}^{-3}$ of ammonium peroxydisulfate.

He recorded the concentration of the peroxydisulfate ions against time and plot the data on the same graph paper.

Based on the data from both experiments on the graph paper, describe how the student can determine that it is first order reaction with respect to iodide ions. [2]

Draw a tangent at time = 0 s for both graphs, the second set of data will give a gradient that has magnitude that is twice of the gradient of the first set of data

- (c) It is reported that Fe^{3+} catalyses the reaction between ammonium peroxodisulfate and iodide ions.

Explain how Fe^{3+} acts as a catalyst. [2]

The catalyst provides an alternative reaction pathway with a lower E_a . Greater proportion of molecules with energy greater than E_a will result in higher frequency of effective collisions. Hence, rate of reaction increases.

- (d) The table below gives the melting points, in $^{\circ}\text{C}$, of the fluorides and chlorides of two elements in Period 3.

	iron	silicon
chlorides	677	-70
iodides	587	120.5

Explain, in terms of structure and bonding, the differences in melting point between

- (i) FeCl_2 and SiCl_4 [2]

FeCl_2 has a giant ionic lattice structure while SiCl_4 has a simple molecular structure

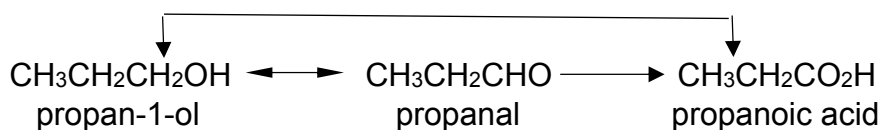
Between Fe^{2+} and Cl^- ions, there are strong ionic bonds while between SiCl_4 molecules there is weak instantaneous dipole induced dipole attractions. Hence more energy is required to overcome the strong ionic bonds in FeCl_2 than the weak instantaneous dipole induced dipole attractions in SiCl_4 .

- (ii) SiI_4 and SiCl_4 [2]

SiI_4 and SiCl_4 has simple molecular structure

SiI_4 has more electrons to be polarised than SiCl_4 , leading to stronger instantaneous dipole induced dipole attractions. Hence more energy required to overcome the instantaneous dipole induced dipole attraction in SiI_4 .

- (e) The Harcourt and Esson reaction is a redox reaction just like the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$. The type of reaction also occurs in organic, as the following transformations from



- (i) State a reagent you could use to convert propanoic acid into propan-1-ol. [1]

LiAlH_4 in dry ether

- (ii) Both propanal and propanoic acid can be obtained from propan-1-ol.

State the reagents you could use to carry these reactions. [2]

$\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, distill for propan-1-ol to oxidise to propanal
 $\text{KMnO}_4(\text{aq})$ or $\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, $\text{H}_2\text{SO}_4(\text{aq})$, heat under reflux for propan-1-ol to oxidise to propanoic acid

- (iii) During the reduction of propanoic acid to propan-1-ol, it was suspected that there are traces of unreacted propanoic acid together with propan-1-ol after the reaction.

Suggest a chemical test to confirm the presence of the unreacted propanoic acid, state the observation that you will see. [2]

Get a small sample of the product, add solid or aqueous sodium carbonate [1]
 effervescence of CO_2 (which gives a white ppt with limewater) will be seen if propanoic acid is present

Reject Na since propan-1-ol will test positive too
 Reject PCl_5 since propan-1-ol will test positive too

- (iv) State the type of reaction that has occurred in (e)(iii). [1]

Acid base reaction

[Total: 20]

- 2 The halogens are a group in the Periodic Table consisting of five chemically related elements: fluorine, chlorine, bromine, iodine and astatine. The group of halogens is the only periodic table group that contains elements in all three main states of matter at standard temperature and pressure.

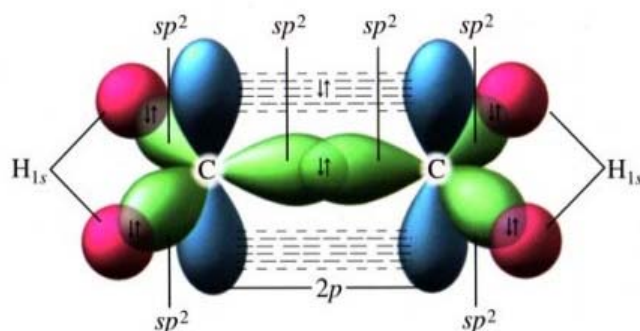
(a) Explain the trend in boiling points of halogens down the group. [2]

The halogens have simple molecular structure with weak instantaneous dipole-induced dipole between the non-polar molecules.

Going down the group, the number of electrons in the halogen molecule increases. The electron cloud becomes more polarisable (i.e. more easily distorted), thus the strength of instantaneous dipole- induced dipole forces increases. More energy is required to overcome the stronger instantaneous dipole- induced dipole forces down the group. This results in an increase in boiling point.

(b) Ethene can react with halogens like chlorine and bromine to form 1,2-dichloroethane and 1,2-dibromoethane respectively.

(i) Describe the bonding between the two carbon atoms in ethene in terms of orbital overlap. You may draw a diagram to illustrate your answer. [2]



The two carbon atoms form double bond (one sigma bond and one pi bond) with each other.

The sp² hybrid orbitals of the carbon atoms overlap head on to form a sigma bond.

The unhybridised p orbitals of the carbon atoms overlap sideways to form a pi bond.

(ii) The enthalpy change of combustion of ethene is given as $-1390 \text{ kJ mol}^{-1}$. Calculate the mass of water at 30°C that can be brought to the boiling point by burning 2.0 dm^3 of ethene.

Assume that 75% of the heat from the ethene is absorbed by the water and volume of gas is measured at room temperature and pressure. [2]

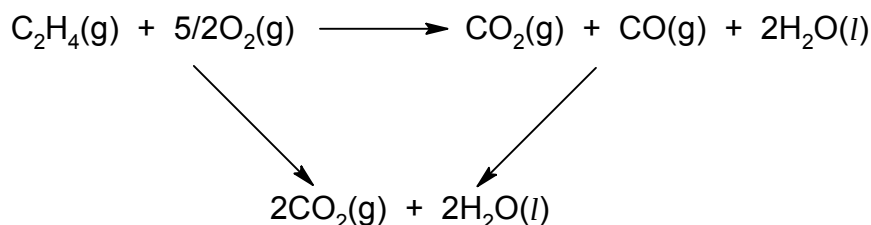
$$\text{Amt of ethene} = 2.0/24 = 0.08333 \text{ mol}$$

$$\text{Heat evolved} = 0.08333 \times 1390 = 115.8 \text{ kJ}$$

Heat gained by water = $115.8 \times 0.75 = 86.88 \text{ kJ}$

Mass of water = $86.88 \times 1000 / (4.18 \times 70) = 297\text{g}$

- (iii) When burnt in a limited supply of air, ethene forms carbon dioxide, carbon monoxide and water according to the equation below.



Calculate the ΔH of the reaction given the energy cycle and following data.

ΔH_c (ethene) = $-1390 \text{ kJ mol}^{-1}$;

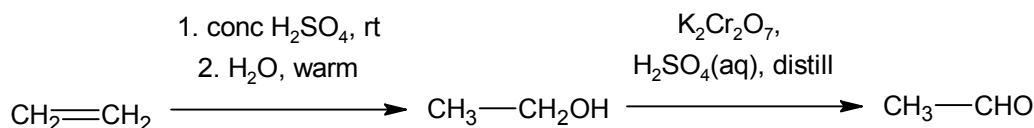
ΔH_f (carbon dioxide) = -394 kJ mol^{-1}

ΔH_c (carbon monoxide) = -283 kJ mol^{-1}

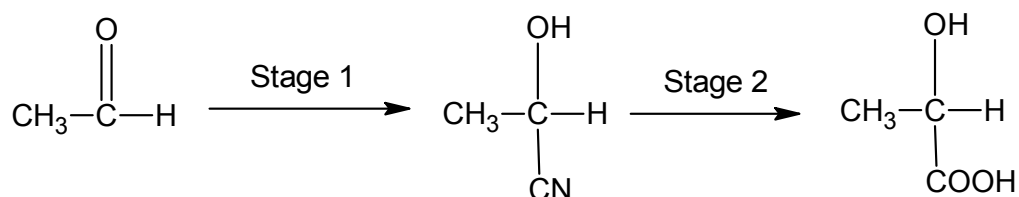
[1]

$$\Delta H = -1390 - (-283) = -1107 = -1110 \text{ kJ mol}^{-1}$$

- (iii) Ethene can be converted to ethanal. State the reagents, conditions needed and the intermediate products formed in the conversion. [3]



- (c) Ethanal undergoes a 3 stage synthesis to form lactic acid as follows.

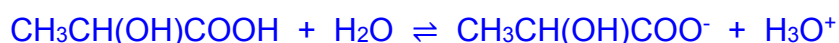


- (i) Name the type of reaction in Stage 1. [1]

Addition

Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ is a monoprotic acid.

- (i) Give the equation which represent the dissociation of lactic acid in water. Identify the acid conjugate base pairs in the equation. [2]



$\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ is an acid. $\text{CH}_3\text{CH}(\text{OH})\text{COO}^-$ is the conjugate base.
 H_2O is a base. H_3O^+ is the conjugate acid of H_2O .

$\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ and $\text{CH}_3\text{CH}(\text{OH})\text{COO}^-$ is a conjugate acid-base pair, while H_2O and H_3O^+ is another conjugate acid-base pair.

- (ii) Explain, using equations, why an aqueous mixture of lactic acid and sodium lactate can act as a buffer solution

(I) on the addition of acid,

(II) on the addition of alkali. [2]

When a small amount of H^+ is added,
 $\text{CH}_3\text{CH}(\text{OH})\text{COO}^- + \text{H}^+ \longrightarrow \text{CH}_3\text{CH}(\text{OH})\text{COOH}$

When a small amount of OH^- is added,
 $\text{CH}_3\text{CH}(\text{OH})\text{COOH} + \text{OH}^- \longrightarrow \text{CH}_3\text{CH}(\text{OH})\text{COO}^- + \text{H}_2\text{O}$

- (e) Lactic acid, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$ is the raw material for the polymer, “polylactide” or PLA.

- (i) What type of reaction takes place in this polymerisation? [1]

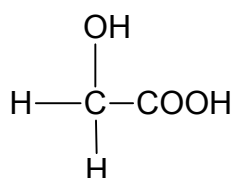
Condensation

One of the reasons PLA has attracted so much attention is that it is biodegradable. This does, however, restrict some potential uses. The simple polymer has a melting point of around 175°C , but softens between $60\text{--}80^\circ\text{C}$. However, its thermoplastic properties enable it to have a range of uses in fibres and in food packaging.

- (ii) Explain why PLA would not be a suitable packing material for foods pickled in vinegar. [1]

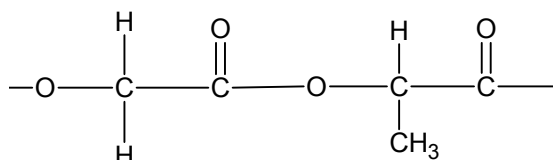
The acid present in vinegar will hydrolyse the ester linkage.

Lactic acid can also be co-polymerised with glycolic acid.



glycolic acid

- (iii) Draw a section of the co-polymer showing one repeat unit. [1]



20

- (iv) Suggest what type of bonding will occur between chains of this co-polymer, indicating the groups involved. [1]

There are van der Waals' forces between methyl groups or permanent dipole-dipole between the ester groups.

[Total: 20]

End of Paper